

GenAI Notes

Module 1: Introduction to Generative AI

1. Fundamentals

What is Artificial Intelligence (AI)?

Definition:

Artificial Intelligence (AI) is the branch of computer science that enables machines to perform tasks that normally require human intelligence.

Examples

- Speech Recognition
- Image Recognition
- Language Translation
- Self-driving Cars
- Virtual Assistants

Goal

Make machines capable of learning, reasoning, problem-solving, and decision-making.

What is Machine Learning (ML)?

Definition:

Machine Learning is a subset of AI where computers learn patterns from data instead of being explicitly programmed.

How it Works

Data → Learn Patterns → Build Model → Make Predictions

Examples

- Email Spam Detection
- Movie Recommendations
- Fraud Detection
- House Price Prediction

Key Idea

The model improves its performance by learning from data.

What is Deep Learning (DL)?

Definition:

Deep Learning is a subset of Machine Learning that uses Artificial Neural Networks with multiple layers to learn complex patterns.

Features

- Learns automatically from large datasets
- Requires high computational power (GPU/TPU)
- Excellent for images, audio, and text

Examples

- Face Recognition
 - Speech Assistants
 - Object Detection
 - Language Translation
-

What is Generative AI?

Definition:

Generative AI is a branch of AI that creates new content by learning patterns from existing data.

It can generate:

- Text
- Images
- Audio
- Video
- Code
- Music
- 3D Objects

Example

Input:

"Generate an image of a cat wearing sunglasses."

Output:

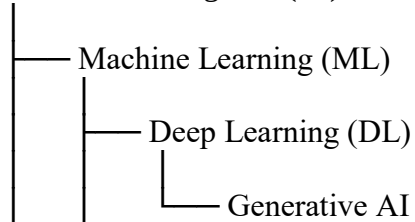
A completely new AI-generated image.

Key Idea

Instead of only predicting or classifying, Generative AI creates new data.

Relationship Between AI, ML, DL, and Generative AI

Artificial Intelligence (AI)



2. Difference between Traditional AI and Generative AI

Traditional AI	Generative AI
Analyzes existing data	Creates new data
Focuses on prediction	Focuses on generation
Performs classification	Produces original content
Rule-based or predictive	Learns data distribution
Output is labels or decisions	Output is text, images, audio, code, etc.

Example

Traditional AI

Image → Cat or Dog?

Generative AI

Prompt → Generate a Cat Image

3. Difference between Discriminative and Generative Models

Discriminative Models

Purpose

Learn the boundary between different classes.

Learn

$P(Y | X)$

Probability of output given input.

Used For

- Classification
- Prediction

Examples

- Logistic Regression
 - SVM
 - Random Forest
 - CNN Classifier
-

Generative Models

Purpose

Learn how data is generated.

Learn

$P(X, Y)$

or the overall data distribution.

Used For

- Content Generation
- Data Synthesis
- Image Creation
- Text Generation

Examples

- GAN

- VAE
 - Diffusion Models
 - GPT
-

Comparison

Discriminative	Generative
Predicts labels	Creates new data
Learns class boundaries	Learns data distribution
Used for classification	Used for generation
Faster training	More computationally expensive
Output is prediction	Output is new content

4. History of Generative AI

1950

- Alan Turing proposed the **Turing Test**.

1956

- AI officially introduced at the Dartmouth Conference.

1980s

- Artificial Neural Networks gained popularity.

1990s

- Probabilistic Generative Models (Bayesian Networks, HMMs).

2013

- Variational Autoencoders (VAEs).

2014

- Generative Adversarial Networks (GANs).

2017

- Transformer architecture introduced.

2020+

- Large Language Models (GPT, Gemini, Claude, etc.) and Diffusion Models revolutionized Generative AI.

5. Evolution of Generative Models

Rule-Based Systems



Statistical Models



Probabilistic Models



Neural Networks



Autoencoders



GANs



Transformers



Diffusion Models



Modern Generative AI

Summary

Generation	Main Idea
Rule-Based	Human-written rules
Statistical	Learn from probabilities
Neural Networks	Learn complex patterns
GANs	Generator vs Discriminator
Transformers	Understand long sequences
Diffusion	Generate high-quality images

6. Applications of Generative AI

Text Generation

- Chatbots
 - Article writing
 - Email drafting
 - Summarization
 - Translation
-

Image Generation

- AI Art
 - Logo Design
 - Product Images
 - Image Editing
-

Code Generation

- Programming assistance
 - Bug Fixing
 - Documentation
 - Unit Test Generation
-

Audio Generation

- Speech Synthesis
 - Voice Cloning
 - Music Generation
-

Video Generation

- AI Videos
 - Animation
 - Video Editing
-

Healthcare

- Drug Discovery
- Medical Reports

- Synthetic Medical Data
-

Education

- Personalized Learning
 - Quiz Generation
 - Tutoring
-

Business

- Marketing Content
 - Customer Support
 - Report Generation
 - Product Design
-

7. Advantages of Generative AI

- Generates content quickly
 - Saves time and cost
 - Increases productivity
 - Automates repetitive tasks
 - Supports creativity
 - Personalized user experiences
 - Works across multiple domains
 - Assists in research and development
-

8. Limitations of Generative AI

- May generate incorrect information (hallucinations)
 - Depends heavily on training data quality
 - High computational cost
 - Can produce biased outputs
 - Limited understanding of real-world context
 - May generate repetitive or inconsistent responses
 - Requires careful prompt design
-

9. Challenges of Generative AI

Technical Challenges

- Large computational requirements
 - High training costs
 - Data quality issues
 - Model scalability
-

Ethical Challenges

- Bias and fairness
 - Misinformation
 - Copyright concerns
 - Deepfakes
 - Privacy issues
-

Business Challenges

- Expensive infrastructure
 - Regulation and compliance
 - Security risks
 - Responsible AI deployment
-

10. Future of Generative AI

Emerging Trends

- More accurate AI models
- Better reasoning capabilities
- Multimodal AI (text, image, audio, video together)
- Personalized AI assistants
- AI-powered software development
- AI in healthcare and education
- Robotics integrated with Generative AI
- Faster and more efficient models
- Responsible and explainable AI
- Greater adoption across industries

Module 4: Autoencoders

1. Introduction

An **Autoencoder (AE)** is a type of **neural network** used for **unsupervised learning**. It learns to **compress input data into a smaller representation** and then **reconstruct the original data** from that compressed representation.

Goal

Learn the important features of data while minimizing information loss.

Example

Input Image → Compress → Reconstruct Same Image

2. What is an Autoencoder?

Definition:

An **Autoencoder** is a neural network that learns to copy its input to its output by passing the data through a compressed representation called the **latent space**.

Input and Output

Input Image → Autoencoder → Reconstructed Image

The output should be as close as possible to the input.

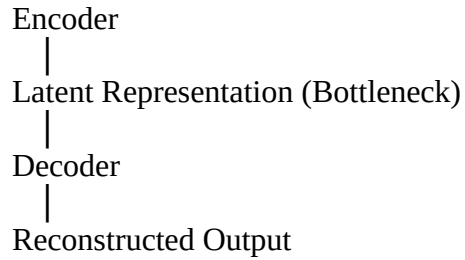
Key Idea

- Learns without labels (Unsupervised Learning)
 - Compresses data
 - Reconstructs data
 - Learns useful features automatically
-

3. Working Principle

The Autoencoder has three main parts:

Input
|



Steps

1. Input data is given.
2. Encoder compresses the data.
3. Bottleneck stores the compressed information.
4. Decoder reconstructs the original input.
5. Compare reconstructed output with original input.
6. Update weights to reduce reconstruction error.

4. Encoder

Definition

The **Encoder** converts high-dimensional input into a **smaller compressed representation**.

Purpose

- Remove unnecessary information.
- Keep only important features.

Example

784 Pixels
↓
256 Neurons
↓
64 Neurons
↓
16 Features

Output

Compressed feature vector.

5. Decoder

Definition

The **Decoder** reconstructs the original input from the compressed representation.

Purpose

Recover the input as accurately as possible.

Example

16 Features
↓
64 Neurons
↓
256 Neurons
↓
784 Pixels

Output

Reconstructed image or data.

6. Bottleneck Layer

Definition

The **Bottleneck Layer** is the smallest hidden layer in the Autoencoder.

Purpose

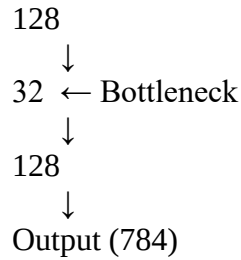
- Forces the model to compress information.
- Keeps only the most important features.

Why is it Important?

Without a bottleneck, the model could simply copy the input instead of learning meaningful patterns.

Example

Input (784)
↓



7. Latent Representation

Definition

The **Latent Representation (Latent Space)** is the compressed encoding produced by the encoder.

It contains the essential features needed to reconstruct the input.

Characteristics

- Lower dimensional
- Information-rich
- Removes redundancy
- Useful for feature extraction

Example

A face image may be represented by features like:

- Face shape
- Eye position
- Nose size
- Smile

instead of storing every pixel.

8. Types of Autoencoders

A. Vanilla Autoencoder

Definition

The simplest Autoencoder with one encoder and one decoder.

Features

- Basic compression
 - Simple architecture
 - Used for reconstruction tasks
-

B. Sparse Autoencoder

Definition

Adds a sparsity constraint so that only a few neurons are active at a time.

Benefits

- Learns meaningful features
 - Reduces redundancy
 - Better feature extraction
-

C. Denoising Autoencoder

Definition

Trained to reconstruct clean data from noisy input.

Process

Original Image
↓
Add Noise
↓
Autoencoder
↓
Clean Image

Application

Image denoising.

D. Contractive Autoencoder

Definition

Adds a penalty that makes the latent representation less sensitive to small input changes.

Benefits

- Learns robust features
 - Better generalization
 - Handles slight input variations
-

E. Convolutional Autoencoder (CAE)

Definition

Uses **Convolutional Neural Networks (CNNs)** instead of fully connected layers.

Used For

- Images
- Videos
- Medical imaging

Advantages

- Preserves spatial information
 - Better image reconstruction
-

F. Stacked Autoencoder

Definition

Multiple Autoencoders stacked together to learn hierarchical features.

Example

Input
↓
Encoder 1
↓
Encoder 2
↓
Encoder 3

↓
Latent Space

Benefits

- Learns deeper representations
 - Improves feature extraction
-

9. Training an Autoencoder

Objective

Train the network so that:

Input $\approx$ Output

Training Steps

1. Input data enters the encoder.
 2. Encoder creates latent representation.
 3. Decoder reconstructs the input.
 4. Calculate reconstruction loss.
 5. Update weights using backpropagation.
 6. Repeat for many epochs.
-

10. Reconstruction Loss

Definition

Measures the difference between the original input and the reconstructed output.

Goal

Minimize reconstruction loss.

Formula

Reconstruction Loss = Difference(Input, Output)

Smaller loss means better reconstruction.

11. Mean Squared Error (MSE) Loss

Definition

Measures the average squared difference between actual and reconstructed values.

Formula

$$\text{MSE} = (1/N) \sum (\text{Actual} - \text{Predicted})^2$$

Characteristics

- Used for continuous data
 - Penalizes large errors more
 - Most common loss for Autoencoders
-

12. Binary Cross Entropy (BCE) Loss

Definition

Measures the difference between actual and predicted probabilities for binary data.

Used When

- Pixel values are normalized between 0 and 1.
- Binary images or binary features.

Characteristics

- Suitable for binary outputs
 - Common in image reconstruction tasks
-

13. Applications of Autoencoders

A. Data Compression

Compress large data into a smaller latent representation while preserving important information.

Example

Large Image → Compressed Features

B. Image Denoising

Remove unwanted noise from images.

Example

Noisy Image → Clean Image

C. Feature Learning

Automatically learn meaningful features from raw data without manual feature engineering.

Used In

- Image recognition
 - Speech recognition
 - NLP
-

D. Anomaly Detection

Train on normal data. If reconstruction error is high, the input is likely abnormal.

Example

Normal Machine Data → Low Error

Faulty Machine Data → High Error → Anomaly Detected

14. Limitations of Autoencoders

- Require large amounts of training data.
- May simply memorize the training data if not designed properly (overfitting).
- Compression can lose important information if the bottleneck is too small.
- Basic Autoencoders are not ideal for generating entirely new data (unlike GANs or VAEs).
- Performance depends on architecture and hyperparameter tuning.
- Training deep Autoencoders can be computationally expensive.

Module 5: Variational Autoencoders (VAE)

1. Fundamentals

A **Variational Autoencoder (VAE)** is a **generative deep learning model** that learns the **probability distribution** of data and generates **new, realistic samples**.

Unlike a traditional Autoencoder, which learns a fixed compressed representation, a VAE learns a **distribution (mean and variance)** for the latent space.

Goal

- Learn the underlying data distribution.
- Generate new data similar to the training data.
- Learn meaningful latent representations.

Example

Face Images → Train VAE → Generate New Face Images

2. Why VAE?

A traditional Autoencoder can only **reconstruct** existing inputs and cannot reliably generate new data.

A **VAE** solves this by learning a **continuous latent distribution**, allowing it to generate new and diverse samples.

Why not a Normal Autoencoder?

Autoencoder	Variational Autoencoder
Learns fixed latent vectors	Learns latent probability distributions
Mainly reconstructs data	Reconstructs and generates new data
Discontinuous latent space	Smooth and continuous latent space

3. Probabilistic Autoencoder

Definition

A **Probabilistic Autoencoder** represents each input as a **probability distribution** instead of a single latent vector.

Instead of learning:

$z = [2.3, 1.8, 0.7]$

The VAE learns:

Mean (μ)

Variance (σ^2)

From these values, it samples a latent vector.

Benefit

- Better generalization
 - Smooth latent space
 - Ability to generate new samples
-

4. Generative Process

A VAE generates data through the following process:

Random Sample (z)



Decoder



Generated Image

Steps

1. Sample a latent vector from the latent distribution.
 2. Pass it to the decoder.
 3. Decoder generates a new sample.
-

5. Latent Distribution

Definition

Instead of storing one latent vector, the encoder predicts a **distribution** characterized by:

- Mean (μ)
- Variance (σ^2)

The latent vector is sampled from this distribution.

Why?

A continuous latent distribution allows smooth transitions and generation of new data.

6. Sampling

Definition

Sampling means selecting a random latent vector from the learned distribution.

Example

Mean = 0

Variance = 1

↓

Random Sample

↓

Decoder

↓

Generated Image

Sampling introduces randomness, enabling different outputs.

7. Reparameterization Trick

Problem

Random sampling is **not differentiable**, so gradients cannot flow through it during training.

Solution

Use the **Reparameterization Trick**:

$$z = \mu + \sigma \times \epsilon$$

where:

- μ = Mean
- σ = Standard deviation
- ε = Random value sampled from $N(0,1)$

Benefit

Allows **Backpropagation** while keeping randomness.

8. Mathematical Concepts

A. Gaussian Distribution

Definition

A **Gaussian (Normal) Distribution** is a bell-shaped probability distribution where most values are near the mean.

Characteristics

- Symmetric
- Bell-shaped
- Defined by Mean and Variance

Example:

```
      *
     * *
    *  *
   *   *
  *     *
```

B. Mean (μ)

Definition

The **Mean** represents the center or average value of the distribution.

Role in VAE

Determines the central location of the latent distribution.

C. Variance (σ^2)

Definition

Variance measures how spread out the data is around the mean.

Interpretation

- Small variance $\rightarrow$ Data is concentrated.
 - Large variance $\rightarrow$ Data is widely spread.
-

D. Standard Normal Distribution

Definition

A Gaussian distribution with:

- Mean = 0
- Variance = 1

Notation:

$N(0,1)$

It is commonly used to sample latent vectors in VAEs.

9. ELBO (Evidence Lower Bound)

Definition

ELBO is the **objective function** that a VAE tries to maximize (or equivalently, minimize its negative).

It balances:

- Reconstruction quality
- Regularization of the latent space

Formula

$ELBO = \text{Reconstruction Loss} + \text{KL Divergence Loss}$

(In practice, training minimizes the negative ELBO.)

Goal

- Reconstruct input accurately.
 - Keep latent distribution close to a standard normal distribution.
-

10. Reconstruction Loss

Definition

Measures the difference between the original input and the reconstructed output.

Goal

Minimize reconstruction error.

Common Loss Functions

- Mean Squared Error (MSE)
- Binary Cross Entropy (BCE)

Smaller reconstruction loss means better reconstruction.

11. KL Divergence Loss

Definition

KL (Kullback–Leibler) Divergence measures how different the learned latent distribution is from the standard normal distribution.

Goal

Make the learned distribution similar to:

$N(0,1)$

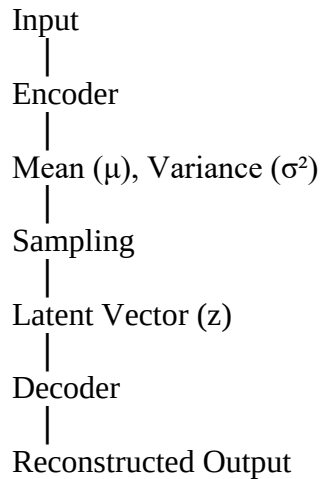
Why is it Important?

- Produces smooth latent space.

- Prevents overfitting.
 - Enables meaningful sampling.
-

12. Architecture

A VAE consists of three main components:



A. Encoder Network

Purpose

- Extract features.
- Predict Mean (μ).
- Predict Variance (σ^2).

Output:

μ, σ^2

B. Latent Space

Definition

A continuous feature space where each point represents a possible data sample.

Benefits

- Smooth interpolation

- Feature learning
 - Data generation
-

C. Decoder Network

Purpose

Generate data from sampled latent vectors.

Input:

Latent Vector (z)

Output:

Generated Image

13. Training

Training Process

1. Input data enters the encoder.
 2. Encoder predicts Mean and Variance.
 3. Sample latent vector using the Reparameterization Trick.
 4. Decoder reconstructs the input.
 5. Compute total loss.
 6. Update weights using Backpropagation.
-

A. Forward Pass

Input



Encoder



μ, σ^2



Sampling



Decoder



Output

B. Sampling

Random latent vectors are generated using:

$$z = \mu + \sigma \times \varepsilon$$

where:

$$\varepsilon \sim N(0,1)$$

C. Loss Optimization

Total Loss:

$$\begin{aligned} \text{Total Loss} = & \\ & \text{Reconstruction Loss} \\ & + \\ & \text{KL Divergence Loss} \end{aligned}$$

Training minimizes this total loss.

14. Applications

A. Image Generation

Generate new realistic images.

Examples:

- Human faces
 - Digits
 - Landscapes
-

B. Image Interpolation

Create smooth transitions between two images by moving through the latent space.

Example:

Face A



Intermediate Faces



Face B

C. Representation Learning

Learn compact and meaningful features for downstream tasks such as classification or clustering.

D. Data Augmentation

Generate additional synthetic training samples to improve model performance.

15. Advantages

- Generates new and realistic data.
 - Learns smooth, continuous latent representations.
 - Good for feature learning.
 - Supports interpolation between samples.
 - Produces diverse outputs.
 - More stable training than GANs.
-

16. Limitations

- Generated images are often blurrier than GAN outputs.
- Balancing Reconstruction Loss and KL Divergence can be challenging.
- Training can be computationally expensive.
- Sensitive to architecture and hyperparameter choices.
- May not capture very fine image details.

Module 6: Generative Adversarial Networks (GAN)

1. Fundamentals

A **Generative Adversarial Network (GAN)** is a **deep learning generative model** that learns to generate new, realistic data by training **two neural networks** against each other.

The two networks are:

- **Generator (G)** → Creates fake data.
- **Discriminator (D)** → Detects whether the data is real or fake.

Goal

Generate data that is so realistic that the discriminator cannot distinguish it from real data.

Example

Random Noise → Generator → Fake Face
↓
Discriminator
Real or Fake?

2. Introduction

GAN was introduced by **Ian Goodfellow** in **2014**.

It is one of the most popular generative models used for:

- Image Generation
- Face Generation
- AI Art
- Image Editing
- Super Resolution

Key Idea

Two neural networks compete with each other and improve over time.

3. History

Year	Development
2014	GAN introduced by Ian Goodfellow
2015	DCGAN
2016	Conditional GAN (CGAN)
2017	CycleGAN, Pix2Pix
2018	StyleGAN, BigGAN
2019	StyleGAN2, ESRRGAN

Year	Development
2021+	StyleGAN3 and improved GAN architectures

4. Why GAN?

Traditional generative models often produce blurry or less realistic outputs.

GANs generate **high-quality and realistic images** because the generator continuously improves by competing against the discriminator.

Advantages over Traditional Models

- Produces realistic images
 - Learns complex data distributions
 - Generates diverse samples
 - No explicit probability calculation required
-

5. Game Theory Basics

GAN training is based on **Game Theory**.

It is a **two-player zero-sum game**.

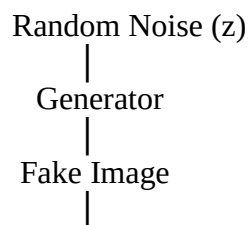
- **Generator** tries to fool the discriminator.
- **Discriminator** tries to correctly identify real and fake data.

Generator ← Competes → Discriminator

If one player improves, the other must improve too.

6. Architecture

A GAN has two neural networks:



Discriminator
|
Real / Fake

A. Generator

Definition

The **Generator (G)** creates fake data from random noise.

Input

Random Noise (Latent Vector)

Output

Generated Image

Goal

Generate data that looks real enough to fool the discriminator.

B. Discriminator

Definition

The **Discriminator (D)** is a binary classifier.

It determines whether the input is:

- Real
- Fake

Goal

Correctly classify real and generated images.

C. Adversarial Training

The Generator and Discriminator are trained together.

- Generator improves by fooling the discriminator.
- Discriminator improves by detecting fake images.

Both networks improve simultaneously.

7. Loss Functions

GAN uses different loss functions for the Generator and Discriminator.

A. Generator Loss

Goal

Generate fake images that the discriminator classifies as **real**.

A lower Generator loss indicates better performance.

B. Discriminator Loss

Goal

Correctly classify:

- Real Images $\rightarrow$ Real
- Fake Images $\rightarrow$ Fake

A lower Discriminator loss indicates better discrimination.

C. Minimax Loss

GAN training is formulated as a **Minimax Game**.

Objective

- Generator minimizes the loss.
- Discriminator maximizes the loss.

This creates a competitive learning process.

8. Training Process

GAN training involves repeatedly updating both networks.

A. Alternating Optimization

Instead of training both networks at once:

1. Train the Discriminator.
 2. Train the Generator.
 3. Repeat.
-

B. Training Loop

Random Noise
↓
Generator
↓
Fake Image
↓
Discriminator
↓
Real/Fake Prediction
↓
Compute Loss
↓
Update Generator & Discriminator

Repeat for many epochs.

C. Convergence

GAN converges when:

- Generator produces realistic images.
- Discriminator cannot reliably distinguish real from fake.

At convergence:

Discriminator Accuracy $\approx 50\%$

This means it is effectively guessing because the generated images look realistic.

9. GAN Problems

A. Mode Collapse

Definition

The Generator produces only a few similar outputs instead of diverse samples.

Example

Instead of generating many different faces, it keeps generating nearly identical faces.

B. Vanishing Gradient

Definition

If the Discriminator becomes too strong, the Generator receives almost no useful gradient to improve.

Result

Generator stops learning.

C. Training Instability

GAN training is sensitive to:

- Learning rate
- Network architecture
- Batch size
- Hyperparameters

Training can easily become unstable.

D. Non-convergence

Sometimes:

- Generator never fools the discriminator.
- Discriminator never reaches balance.

The model fails to converge to a stable solution.

Module 7: GAN Variants

Different GAN variants were developed to solve specific problems and improve performance.

1. DCGAN (Deep Convolutional GAN)

Definition

A GAN that replaces fully connected layers with **Convolutional Neural Networks (CNNs)**.

Architecture

Noise
↓
CNN Generator
↓
Generated Image
↓
CNN Discriminator
↓
Real/Fake

Advantages

- Better image quality
 - Stable training
 - Learns image features automatically
-

CNN Generator

Uses:

- Transposed Convolutions (Deconvolutions)
- Batch Normalization
- ReLU activation (mostly)

Purpose:

Generate realistic images from noise.

CNN Discriminator

Uses:

- Convolution Layers
- Leaky ReLU activation
- Downsampling

Purpose:

Classify images as real or fake.

2. Conditional GAN (CGAN)

Definition

A Conditional GAN generates images based on additional input called a **condition**.

The condition can be:

- Class label
- Text
- Image
- Attribute

Example

Noise + Label ("Cat")
↓
Generator
↓
Generated Cat Image

Applications

- Digit generation
 - Text-to-image generation
 - Controlled image generation
-

Conditional Generation

Instead of generating random images, the Generator creates images that satisfy the given condition.

Example:

Input Label = Dog

↓

Generated Dog Image

3. Pix2Pix

Definition

Pix2Pix is a GAN designed for **paired Image-to-Image Translation**.

It learns a mapping between corresponding input and output images.

Examples

- Sketch → Photo
 - Black & White → Color
 - Satellite → Map
-

Image-to-Image Translation

Convert one image into another while preserving structure.

Example

Sketch

↓

Pix2Pix



Realistic Face

4. CycleGAN

Definition

CycleGAN performs **Image-to-Image Translation without paired training data**.

Example

- Horse ↔ Zebra
- Summer ↔ Winter
- Painting ↔ Photo

No matching image pairs are required.

Unpaired Translation

Uses separate collections of images instead of one-to-one pairs.

Example:

Horse Dataset



CycleGAN



Zebra Images

Cycle Consistency Loss

Ensures the translated image can be converted back to the original image.

Example:

Horse



Zebra



Horse

The reconstructed horse should closely match the original.

5. WGAN (Wasserstein GAN)

Definition

WGAN improves GAN training by replacing the original GAN loss with the **Wasserstein Distance**.

Benefits

- More stable training
 - Better convergence
 - Reduced Mode Collapse
-

Wasserstein Distance

Measures the distance between the real and generated data distributions.

It provides a smoother and more informative training signal than the original GAN loss.

Earth Mover Distance (EMD)

Another name for the **Wasserstein Distance**.

Intuition

Imagine moving piles of earth to transform one distribution into another.

The minimum work required is the Earth Mover Distance.

Smaller distance = Generated distribution is closer to the real distribution.

6. WGAN-GP (WGAN with Gradient Penalty)

Definition

An improved version of WGAN that replaces **weight clipping** with a **Gradient Penalty**.

Gradient Penalty

Adds a penalty term during training to enforce smooth gradients.

Advantages

- More stable than WGAN
 - Faster convergence
 - Better image quality
-

7. StyleGAN

Definition

StyleGAN generates highly realistic images by controlling image styles at different levels.

Widely used for realistic face generation.

Style Mapping

A **Mapping Network** converts the input noise into a style vector.

This style controls features such as:

- Hair color
 - Face shape
 - Smile
 - Age
-

Adaptive Instance Normalization (AdaIN)

AdaIN injects style information into each layer of the Generator.

Benefit

Enables fine control over image appearance.

Progressive Growing

Training starts with **low-resolution images** and gradually increases the image resolution.

Advantages

- Stable training
 - High-quality image generation
 - Better fine details
-

8. BigGAN

Definition

BigGAN is a large-scale GAN trained on massive datasets with larger models.

Features

- High-resolution image generation
- Large batch training
- Class-conditional image generation

Applications

- Image synthesis
 - Research
 - Benchmarking
-

9. SRGAN (Super-Resolution GAN)

Definition

SRGAN converts **low-resolution images** into **high-resolution images**.

Example

Low Resolution Image



SRGAN



High Resolution Image

Applications

- Medical imaging
 - Satellite imagery
 - Photo enhancement
-

10. ESRGAN (Enhanced SRGAN)

Definition

ESRGAN is an improved version of SRGAN that generates sharper and more realistic high-resolution images.

Improvements over SRGAN

- Better texture reconstruction
- Sharper details
- More stable training

Module 8: Diffusion Models

1. Fundamentals

A **Diffusion Model** is a **generative AI model** that learns to create new data by **gradually removing noise** from random noise.

Unlike GANs that generate data in one step, Diffusion Models generate data through **multiple denoising steps**.

Goal

- Learn how to reverse the process of adding noise.
- Generate realistic images from random noise.

Example

Random Noise



Gradually Remove Noise



Realistic Image

2. Introduction

Diffusion Models are inspired by the physical process of **diffusion**, where particles spread over time.

In AI:

- **Forward Process** → Gradually adds noise to data.
- **Reverse Process** → Learns to remove noise and recover the original data.

Today, Diffusion Models power systems like **Stable Diffusion** and many modern AI image generators.

3. Why Diffusion Models?

GANs often suffer from:

- Mode Collapse
- Training Instability
- Difficult optimization

Diffusion Models solve many of these problems.

Advantages over GANs

- Stable training
 - High-quality image generation
 - Diverse outputs
 - Better coverage of data distribution
-

4. Intuition

Imagine gradually covering a clear image with random noise until it becomes completely random.

Now imagine learning to reverse this process.

Original Image



Small Noise



More Noise



Complete Noise

The model learns to go in the opposite direction:

Noise



Less Noise



Cleaner Image



Original Image

5. Forward Diffusion

Definition

Forward Diffusion is the process of **gradually adding Gaussian noise** to the original data over many steps.

Eventually, the image becomes pure random noise.

Process

Original Image



Add Noise



More Noise



Pure Gaussian Noise

The forward process is fixed and does not require learning.

6. Noise Addition

At each timestep, a small amount of random noise is added to the image.

This continues until almost all image information is destroyed.

Purpose

Create training examples for the reverse denoising process.

7. Gaussian Noise

Definition

Gaussian Noise is random noise sampled from a **Normal (Gaussian) Distribution**.

Notation:

$N(0,1)$

where:

- Mean = 0
- Variance = 1

Why Gaussian Noise?

- Easy to sample
 - Mathematically simple
 - Produces smooth learning
-

8. Markov Chain

Definition

The forward diffusion process is modeled as a **Markov Chain**.

A Markov Chain means:

The current state depends only on the previous state, not on earlier states.

Example

Image₀
↓
Image₁
↓
Image₂

↓
Image₃

Each noisy image depends only on the previous noisy image.

9. Reverse Diffusion

Definition

Reverse Diffusion is the process of **gradually removing noise** from a noisy image to recover a clean image.

Unlike the forward process, the reverse process is **learned** by a neural network.

Process

Random Noise
↓
Less Noise
↓
Cleaner Image
↓
Generated Image

10. Denoising

Definition

Denoising means predicting and removing the noise present in the image.

Instead of predicting the final image directly, the model predicts the **noise** contained in the image.

11. Noise Prediction

The neural network learns:

Noisy Image
↓
Predict Noise
↓

Remove Noise



Cleaner Image

Predicting noise is easier and leads to more stable training.

12. Score-Based Learning

Definition

Score-Based Learning teaches the model the **direction in which noise should be removed**.

The model learns how to move noisy samples toward regions with higher probability of real data.

Benefit

- Better denoising
 - Better image quality
-

13. Diffusion Objective

Goal

Train the model to accurately predict the added noise.

The objective is to minimize the difference between:

- Actual noise
- Predicted noise

Lower prediction error results in better image generation.

14. Sampling

Sampling is the process of generating a new image.

Steps

Random Noise



Reverse Diffusion



Generated Image

The model repeatedly removes noise until a realistic image is obtained.

15. DDPM (Denoising Diffusion Probabilistic Model)

Definition

DDPM is the original and most common Diffusion Model.

Features

- Uses many denoising steps (often around 1000).
 - Produces very high-quality images.
 - Slower image generation due to many iterations.
-

16. DDIM (Denoising Diffusion Implicit Model)

Definition

DDIM is an improved version of DDPM.

Advantages

- Faster image generation
 - Fewer sampling steps
 - Similar image quality
-

17. Noise Scheduler

Definition

A Noise Scheduler controls **how much noise is added or removed at each timestep**.

Purpose

- Improve training stability
- Improve sampling quality

- Control the diffusion process
-

18. Timesteps

Definition

Timesteps represent the number of diffusion steps.

Example:

$t = 0$

Original Image

↓

$t = 1000$

Complete Noise

More timesteps generally lead to better quality but slower generation.

19. Classifier Guidance

Definition

Classifier Guidance uses a **separate classifier** to guide image generation toward a desired class.

Example

Prompt:

Generate a Dog

The classifier encourages the model to generate a dog image.

Limitation

Requires an additional classifier model.

20. Classifier-Free Guidance (CFG)

Definition

Classifier-Free Guidance guides generation **without requiring a separate classifier**.

Instead, the diffusion model learns both:

- Conditional generation
- Unconditional generation

Advantages

- Better image quality
- Faster inference
- Simpler architecture

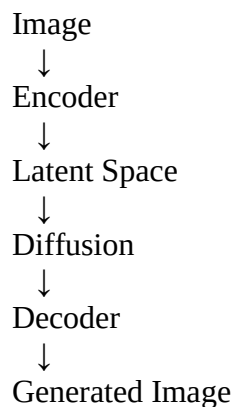
It is widely used in **Stable Diffusion**.

21. Latent Diffusion

Definition

Instead of performing diffusion directly on images, **Latent Diffusion Models (LDMs)** perform diffusion in a **compressed latent space**.

Process



Advantages

- Faster generation
- Lower memory usage
- Reduced computational cost

Stable Diffusion is based on Latent Diffusion.

22. Applications

A. Image Generation

Generate realistic images from text prompts or random noise.

Examples:

- AI Art
 - Human Faces
 - Landscapes
 - Product Images
-

B. Image Editing

Modify existing images.

Examples:

- Change background
 - Change color
 - Remove objects
 - Add objects
-

C. Inpainting

Definition

Fill in missing or damaged parts of an image.

Example

Broken Image



Diffusion Model



Restored Image

D. Outpainting

Definition

Extend an image beyond its original boundaries.

Example

Original Image



Larger Image with New Background

E. Super Resolution

Increase the resolution of low-quality images while preserving details.

Example

Low Resolution



Diffusion Model



High Resolution

23. Advantages

- Generates highly realistic images.
 - Stable training compared to GANs.
 - Produces diverse outputs.
 - Supports text-to-image generation.
 - Excellent image editing capabilities.
 - Works well for inpainting and outpainting.
 - Scales effectively to large datasets.
-

24. Limitations

- Slow image generation due to many denoising steps.
- High computational and memory requirements.
- Expensive to train.
- Requires powerful GPUs for large models.

- Sampling can take many iterations.

Module 18: Multimodal Generative AI (Without LLMs)

Note: These concepts focus on **Multimodal Generative AI without Large Language Models (LLMs)**. The emphasis is on how different data types (modalities) work together using traditional deep learning and generative models.

1. Multimodal Learning

Definition

Multimodal Learning is a machine learning approach where a model learns from **multiple types of data (modalities)** simultaneously.

A **modality** is a type of data such as:

- Text
- Image
- Audio
- Video
- Sensor Data

Goal

Learn relationships between different types of data to improve understanding and generation.

Example

Image + Text → Learn Relationship

or

Audio + Video → Learn Together

Benefits

- Better understanding of data
 - More accurate predictions
 - Richer content generation
 - Handles complex real-world tasks
-

2. Vision-Language Alignment (Conceptual)

Definition

Vision-Language Alignment is the process of learning the relationship between **images and text**.

The model learns that certain visual objects correspond to specific words or descriptions.

Example

Image of a Dog

↔

"The dog is running in a park."

The model learns that:

- Dog → Animal
- Running → Action
- Park → Scene

Goal

Map images and text into a common feature space where related image-text pairs are close together.

Applications

- Image Search
 - Image Captioning
 - Visual Question Answering
 - Cross-modal retrieval
-

3. Image Captioning (Traditional)

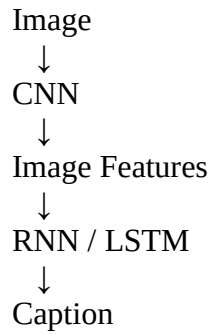
Definition

Image Captioning is the task of automatically generating a textual description of an image.

Traditional image captioning systems typically use:

- **CNN** → Extract image features
- **RNN/LSTM/GRU** → Generate the sentence

Workflow



Example

Input:

□ Image of a cat sitting on a sofa.

Output:

"A cat is sitting on a sofa."

Applications

- Assistive technology
- Photo organization
- Accessibility
- Digital libraries

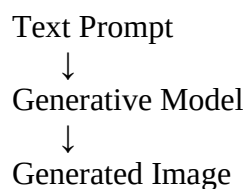
4. Text-to-Image (Concept)

Definition

Text-to-Image Generation creates an image from a text description.

The model learns the relationship between text and images during training.

Workflow



Example

Input:

"A red sports car on a mountain road."

Output:

A newly generated image matching the description.

Traditional Models Used

- GANs
- VAEs
- Diffusion Models

(Modern systems often use LLMs with diffusion models, but the core concept exists without LLMs.)

Applications

- AI Art
 - Graphic Design
 - Product Design
 - Content Creation
-

5. Image-to-Image

Definition

Image-to-Image Translation transforms one image into another while preserving important content.

The input and output are both images.

Workflow

Input Image
↓
Generative Model
↓
Output Image

Examples

- Black & White → Color
- Sketch → Realistic Photo
- Day → Night
- Horse → Zebra
- Low Resolution → High Resolution

Popular Models

- Pix2Pix
- CycleGAN
- Diffusion Models

Applications

- Photo Editing
 - Medical Imaging
 - Style Transfer
 - Image Enhancement
-

6. Image-to-Audio

Definition

Image-to-Audio Generation converts visual information into corresponding audio.

The generated audio may be:

- Speech
- Sound Effects
- Music
- Environmental Sounds

Workflow

Image
↓
Feature Extraction
↓
Audio Generation

Example

Input:

Prathamesh Arvind Jadhav

☐ Image of rainfall.

Output:

☐ Rain sound.

Another Example:

☐ Image of a piano.

Output:

☐ Piano music.

Applications

- Accessibility for visually impaired users
- Multimedia generation
- Virtual Reality
- Gaming

7. Cross-Modal Generation

Definition

Cross-Modal Generation is the process of generating data in **one modality** using information from **another modality**.

Examples

Text → Image

Image → Text

Image → Audio

Audio → Image

Video → Text

Goal

Learn relationships between different data types and generate meaningful outputs across modalities.

Applications

- Image Captioning

- Text-to-Image Generation
- Speech-to-Image
- Audio Description
- Multimedia Content Creation